

Half & Full Duplex communication

```
pc---pc           Full Duplex
pc---switch       Full Duplex
pc---router       Full Duplex
pc---Hub          Half Duplex
Switch---hub      Half Duplex
Router---Switch   Full Duplex
```

Default Routing:

1. Default routing is used in a stub nw. A network with one entry/exit point.
 2. Default routing is used, when you are working with multiple unknown networks.
 3. When Default routing is configured, if router receives any unknown packet(whose entry is not available in the routing table, router will simply transfer/forward the packets towards exit interface, so that it can be delivered to connected ISP.
 4. Special kind of static Routing.
 5. Default routing is used, when destination networks are unknown, & need to be redirected to other router(instead of dropping an unknown packet).
3. represented by S*

4.command to configure Default Routing:

- 1.using next hop ip:

```
=====
config# ip route 0.0.0.0 0.0.0.0 1.1.1.1
#do show ip route
```

OR

- 2.using self exit:

```
=====
config# ip route 0.0.0.0 0.0.0.0 s0/0
#do show ip route
```

Default Routing can be done in 2 ways

1. self exit
2. Next hop

Banners show a welcome message to the connected users

1. motd: Message of the day
2. login:

```
config# enable secret sec@123
line vty 0 4
password 123
login
exit
config# banner motd # Have a nice day #
config# end          to go to privilege mode
config# exit         to go to user execution mode
```

```
config# banner login # Hope, you are going to enjoy our services #
```

Types of Router/Switch Passwords

1. Normal

no enable password

to secure privilege mode

```
config# enable password nor@123
```

```
exit
exit
>enable
password:
```

```

2. Secret/encrypted                to secure privilege mode
config# enable secret
sec@123                            no enable secret
exit
exit
>enable
password:

3. Console                        to secure console mode (user exec mode
pwd)
config# line conole 0
config-line)#password con@123
no password
no login
#login
#end
#exit
password:

4. Aux                            to secure aux mode
config# line aux
0
config-line)#password
aux@123
#login
#end
#exit

5. Telnet                        to protect telnet access
config# line vty
0 4
config-line)#password
tel@123
#login
#end
goto any pc---cmd
>telnet
192.168.1.1
password

```

To manually encrypt passwords (normal, console, aux, telnet):

```

config# service password-encryption
#end
#sh run
to verify

```

=====

Startup-config backup & restore (saved configuration)

1. connect a server machine to Router(10.0.0.1/8)
2. assign ip of same nw(eg:10.0.0.2/8)

3. ping server with router
4. on Router: to take backup
 - hostname Mumbai
 - Mumbai#copy run start
 - #copy start tftp

```

10.0.0.2
press enter(to use default filename)

```

To restore:

```
#erase start          to delete saved configuration
#copy tftp start
10.0.0.2
Mumbai-config
enter (to continue with default name)
```

=====

=

Router password Recovery:

```
router          config# enable secret sec@123
                #end
                #write

                #reload
                #enable
                it will ask for password
```

to recover the pwd:

```
goto physical tab---power off/on
```

```
goto cli tab---ctrl+c
```

```
rommon1>?
```

```
rommon2>confreg 0x2142
```

to change configuration register

value for password

recovery(this will skip the step of reading nvram)

```
rommon3>boot
```

```
Basic Setup ..... n
```

```
Router>en
```

```
config# copy start run
#no enable secret
```

to remove

pwd

```
#config-register 0x2102
```

to set the configuration

register value back to normal mode.

```
#end
```

```
#write
```

```
#reload
```

No password will be asked for privileged mode.